Research article

The lonely algorithm problem.

Samuel Hardman Taylor
University of Illinois Chicago

Y. Anthony Chen
Stanford University

Personalization promises feeds that feel made for us, yet many people report feeling more alone the longer they scroll. This study asks how perceptions of algorithmic personalization relate to social connectedness over time. Using a three-wave panel of adult TikTok users, we test whether feeling understood by the algorithm substitutes for, or supplements, feeling understood by other people.

Keywords algorithms · loneliness · social connectedness · TikTok · panel study

Introduction

Recommendation algorithms now sit between people and much of what they see, read and share. Users describe their feeds in strikingly relational terms: the algorithm "gets" them, "knows" them, or has stopped listening. These folk descriptions invite a communication question. If people experience algorithms as responsive, what happens to their sense of being understood by the people around them?

Research on perceived responsiveness has long shown that feeling understood, validated and cared for is a cornerstone of close relationships. We extend this work to algorithmic media, asking whether a responsive feed offers some of the same rewards, and at what cost.

Two accounts compete. A displacement account suggests that time with responsive feeds crowds out time with people. A stimulation account suggests that feeling seen by an algorithm builds confidence that spills over into offline ties.

Theoretical framework

We draw on the perceived algorithm responsiveness framework, which distinguishes feeling understood by a system from simply receiving relevant content. The distinction matters because understanding is relational: it implies a partner who notices and adjusts.

Method

We recruited adult TikTok users through a national online panel and surveyed them three times, six weeks apart. At each wave participants reported how responsive their feed felt, how connected they felt to friends and family, and how much time they spent on the platform each day.

Perceived responsiveness was measured with an established scale capturing feeling understood, validated and cared for by the feed. Social connectedness was measured with a widely used belonging scale. We estimated random-intercept cross-lagged panel models to separate stable differences between people from changes within each person over time.

Results

Participants who reported higher perceived algorithm responsiveness at one wave reported slightly lower social connectedness at the next, even after accounting for prior connectedness and time spent on the platform. The reverse path was weaker but also present.



Figure 2. Cross-lagged associations between perceived responsiveness and social connectedness across three waves (sample data).

These patterns held across age groups and were strongest for participants who used the platform late at night, when other sources of connection were least available.

Discussion

Our findings suggest that algorithmic responsiveness behaves like a partial substitute for interpersonal responsiveness. It satisfies something, briefly, while making the harder work of human connection feel less urgent.

For designers, this points to features that hand people back to one another. For researchers, it argues for treating algorithms as communicative partners whose effects unfold over weeks, not minutes.

Limitations

The sample was limited to adult users in the United States, and self-reports of responsiveness may blend perceptions of content quality with perceptions of understanding.

Conclusion

Algorithms are becoming one of the places where people look for the feeling of being known. Understanding when that search supports human connection, and when it quietly replaces it, is a central task for communication research.